

Surface Rings in Euclidean Space

An Accessible Technical Hook with Planar/Spherical Examples

Prepared for physicists and applied theorists

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Abstract

This short note gives a hands-on view of Surface Rings (S-rings) in familiar Euclidean settings. We derive the split-skew boundary law on a plane and on a sphere, show how the wavenumber operator $K = \sqrt{-\Delta_S}$ appears in each case, and outline two worked example tracks: (i) photon/scalar propagation across a flat interface and (ii) hydrogenic integrals using momentum-space weights. We also sketch the spin-2 (graviton) limit and the $1/r$ potential recovery. The aim is a practical bridge from the formal canon to calculations a physicist or advanced engineer can run.

1 A narrative picture before the math

Think of a surface S separating two sides of a medium (or two models of the same world). Every field you care about has two traces on S : what is *shared* across the surface (consensus) and what *differs* (contrast). S-rings take this split seriously and say: the fluxes that cross S are the sum of a *persistence* piece (causal, passive, same on both slots) and a *handed rotation* between consensus and contrast (purely lossless, flips direction when you flip the normal).

That one sentence already encodes the essential physics:

- **Two slots, one law.** You never need more than “difference” and “average” on the boundary, and the law is linear in that 2D space.
- **Causality and passivity.** The even part of the law is a Herglotz/Stieltjes kernel: it can be written as positive spectral measures in frequency and in the surface operator K^2 . No ad-hoc knees are required—thresholds and relaxation become *measures*.
- **Direction without dissipation.** The skew (odd) part is a clean 90° rotation between the two slots; it stores no energy and simply records the sense of “outward”. Flipping the normal flips this sign.
- **Rings and surfeons.** When you look in (ω, K) -space, there are lossless bands (“rings”) where dissipation vanishes. Quantizing a mode on such a band gives you a particle-like boundary excitation: a *surfeon*. Chains of rings give you composites.

Once you accept this boundary-first view, Euclidean calculations are straightforward: on a plane, K is just tangential Fourier magnitude; on a sphere, K is the square-root of the Laplace–Beltrami eigenvalue, $\kappa_\ell = \sqrt{\ell(\ell+1)}/R$. Everything else is plugging your favorite $Z(\omega, K)$ (as a positive spectral measure) into familiar integrals.

2 Set-up and split-skew canon

Let S be a smooth oriented surface in $\mathbb{R}^3$. For a bulk field q with conjugate flux p (power density $\langle p, \dot{q} \rangle_S$), fold boundary traces across S as

$$\Psi := \begin{pmatrix} [q] \\ \{q\} \end{pmatrix}, \quad \Theta := \begin{pmatrix} [p] \\ \{p\} \end{pmatrix}, \quad [f] = f_+ - f_-, \quad \{f\} = (f_+ + f_-)/2.$$

Under causality, passivity, tangential stationarity, and local isotropy, each polarization/irrep channel obeys the split-skew boundary law

$$\Theta(\omega) = [Z(\omega, K) I + s(K) J] \Psi(\omega), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (1)$$

with $K := \sqrt{-\Delta_S}$. Here Z is *Herglotz* in $i\omega$ and *Stieltjes* in K^2 (admits positive spectral measures), and $s(K)$ is real, lossless, and flips under normal reversal ($\hat{n} \rightarrow -\hat{n}$). Equation (1) is the Euclidean-face of the representation theorem proved in the companion formal note.

Planar S : $z = 0$ (Fourier along x, y). For a scalar Helmholtz bulk $(-\Delta - k_0^2)\phi = 0$ in $z > 0$, $z < 0$, a tangential Fourier mode $e^{i\xi \cdot x}$ decays as $e^{-\gamma z}$ with $\gamma(\xi) = \sqrt{k_0^2 + |\xi|^2}$ (branch $\Re \gamma > 0$). The Dirichlet-to-Neumann (DtN) operator is a multiplier: $\widehat{\partial_n \phi}(\xi, 0) = \gamma(\xi) \hat{\phi}(\xi, 0)$. Identifying $K \leftrightarrow |\xi|$ gives $Z(\omega, K) = \gamma(K)$, a complete Bernstein/Stieltjes function of K^2 . In reciprocal media $s \equiv 0$; magnetoelectric skew yields $s \neq 0$ as a lossless add-on.

Spherical S : $|\mathbf{r}| = R$ (spherical harmonics). The Laplace–Beltrami eigenfunctions are $Y_{\ell m}$ with $-\Delta_S Y_{\ell m} = \ell(\ell + 1)R^{-2}Y_{\ell m}$. Hence K has eigenvalues $\kappa_\ell = \sqrt{\ell(\ell + 1)}/R \sim \ell/R$ for large ℓ . Any K -dependent kernel acts diagonally in $\{Y_{\ell m}\}$: $Z(\omega, K) \rightarrow Z(\omega, \kappa_\ell)$. Isotropy per channel permits lossless skew $s(\kappa_\ell)$ that flips with the outward normal.

3 Photon/scalar across a flat interface

Scalar. On $z = 0$, the DtN symbol $\gamma(K) = \sqrt{k_0^2 + K^2}$ gives the even part of the S-ring kernel: $\Xi(K) := \Re Z(0, K) = \gamma(K)$. A simple dispersion $\omega^2 = c^2(K^2 + k_0^2)$ follows from continuity of tangential phase. Knee-free modeling: use a (finite) Stieltjes sum $\Xi(K) = a_0 + \sum_j a_j/(U_j + K^2)$ with $a_j \geq 0$, or a positive density $\int \rho(u)/(u + K^2) du$.

Electromagnetism (TE/TM). For index $n(\omega)$, $k_0 = \omega/c$, the surface admittances are $Y_{\text{TE}} = \gamma/(i\omega\mu)$, $Y_{\text{TM}} = i\omega\varepsilon/\gamma$, Herglotz in $i\omega$ and Stieltjes in K^2 . After power-normalization, each polarization fits (1); optional magnetoelectric skew contributes $s(K)J$.

4 Hydrogenic integrals in momentum space

Momentum-space weights for bound states are spherical Bessel transforms of radial wavefunctions. In atomic units ($a_0 = 1$),

$$R_{10}(r) = 2e^{-r}, \quad R_{20}(r) = \frac{1}{2\sqrt{2}}(2 - r)e^{-r/2}, \quad R_{21}(r) = \frac{1}{2\sqrt{6}}re^{-r/2}. \quad (2)$$

The (spherically averaged) momentum distribution for angular momentum ℓ is

$$P_{n\ell}(k) \propto k^2 \left| \int_0^\infty r^2 R_{n\ell}(r) j_\ell(kr) dr \right|^2, \quad \int_0^\infty P_{n\ell}(k) dk = 1,$$

with j_ℓ spherical Bessel. Define the hydrogenic contrast $W_{2S}-W_{2P}$ and the 1S weight W_{1S} . Then the leading S-ring energy shifts are linear functionals of the same static kernel $\Xi(K) = \Re Z(0, K)$:

$$\Delta E_{2S \rightarrow 2P} = C \int_0^\infty \Xi(K) [W_{2S}(K) - W_{2P}(K)] dK, \quad \Delta E_{1S} = C \int_0^\infty \Xi(K) W_{1S}(K) dK. \quad (3)$$

Calibration: fix the single normalization C on a robust centroid (e.g., FOSOF $2S \rightarrow 2P$). If the 1S level demands high- K suppression, insert a positive UV spectral density in the Stieltjes representation of Ξ with a Compton-scale threshold; this preserves passivity/causality and leaves the IR constants (and GR) untouched.

Numerical recipe (outline).

- Precompute $W_{n\ell}(K)$ via fast 1D quadrature of $R_{n\ell} * j_\ell$; normalize $\int W dK = 1$.
- Choose a knee-free $\Xi(K) = a_0 + \sum_j a_j / (U_j + K^2)$ with $a_j \geq 0$ (or a positive density); fit $a_\bullet$ to IR constraints (GR) and the centroid via C .
- Evaluate the two integrals; adjust only the UV density (not IR) if 1S needs suppression.

5 Spin-2 (graviton): from waves to Newton

What the SR graviton is

On an S-ring, the transverse–traceless (TT) edge metric perturbation h_{ab}^{TT} has two polarizations $\lambda \in \{+, \times\}$, and the boundary law in the linear regime reads schematically

$$[K_{ab} - K \gamma_{ab}]_\lambda^{\text{TT}} = 8\pi G \left(Y^{(2)}(\omega, K) I + s^{(2)}(K) J \right) [h^{\text{TT}}]_\lambda. \quad (4)$$

The *same* static kernel $\Xi(K) := \Re Z(0, K)$ controls the small- K spin-2 admittance, $Y^{(2)}(\omega, K) \propto \Xi(K)$ as $K \rightarrow 0$, while $s^{(2)}(K)$ is a tiny, lossless parity-odd term that flips under normal reversal. Quantizing the TT field on a lossless band gives *spin-2 surfеons* (gravitons) carrying $+, \times$.

Coupling and equivalence principle (edge view)

Varying the SR action with respect to the boundary metric yields the edge stress T_{edge}^{ab} . The universal spin-2 admittance implies the minimal coupling

$$\mathcal{L}_{\text{int}}^{(2)} = \frac{1}{2} h_{ab}^{\text{TT}} T_{\text{edge}}^{ab}, \quad (5)$$

independent of composition: any surfеon chain couples through the same Ξ (equivalence principle at the boundary). Inertial loading, on the other hand, is set by the $K \rightarrow 0$ offset of the chain (the share in ε_0). Because both descend from Ξ , $m_{\text{grav}} = m_{\text{in}}$ is automatic in the SR frame.

From spin-2 waves to Newton’s $1/r$

In the static limit, the boundary Green’s function induced by Ξ produces the familiar massless propagator in the bulk,

$$V(r) \propto \int \frac{d^3 q}{(2\pi)^3} \frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{q^2} = \frac{1}{4\pi r}, \quad (6)$$

so, after normalization to Cassini/Shapiro (PPN γ) data, one recovers $V(r) = -GM/r$. The Cassini bound $\max_b |\gamma - 1| \leq 2.3 \times 10^{-5}$ simply constrains how flat $\Xi(K)$ must be near $K=0$; any parity-odd $s^{(2)}(K)$ is then correspondingly tiny, with a distinctive *normal-flip* signature in polarization-sensitive GW analyses.

6 Quick start (Euclidean track)

- **Plane** $z = 0$: use $\gamma(K) = \sqrt{k_0^2 + K^2}$, optional $s(K)$; validate reflection/transmission and power.
- **Sphere** $|\mathbf{r}| = R$: diagonalize in $Y_{\ell m}$; choose $\Xi(\kappa_\ell)$ knee-free; check automode bands (where $\Re Z = 0$).
- **Hydrogen**: compute $W_{2S} - W_{2P}$ and W_{1S} ; fit centroid with C ; test 1S with a positive UV density.
- **Spin-2**: verify $1/r$ via the static transform and constrain the small- K neighborhood of Ξ by Cassini's γ .

Pointers. Formal proofs (Herglotz/Stieltjes, Dirichlet forms, DtN/Calderón, port-Hamiltonian) and extended derivations: see `Research Journal/SR.Formalization_and_Validations.tex`. Front narrative and surfeon framing: `Research Journal/Surfeon_Frontend.tex`.